SPCC: Star-based Private Clique Counting

Anonymous Author(s)
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Abstract

Clique counting is a fundamental task in graph analysis with many real-world applications, such as social network and cybersecurity analysis. However, directly releasing these statistics raises privacy concerns, as many graphs are derived from personal data. Local Differential Privacy (LDP) is a promising privacy model, but existing LDP methods for clique counting are not fully feasible, as they primarily rely on *edge-level perturbation*, suffering from excessive noise and high communication overhead, especially for higher-order cliques. This paper introduces **SPCC**, a novel LDP framework that shifts the data representation and perturbation unit from individual edges to cohesive k-**stars**. A k-star, defined as a center node connected to k neighbors, naturally serves as the structural building block for cliques and captures local topology. To perturb k-star sets, SPCC incorporates two key mechanisms: (i) *Padding-and-Dropping (PD)* mechanism that standardizes report lengths with a theoretically derived optimal value L to conceal true degrees and minimize error; and (ii) *Adaptive Grouping Perturbation (AGP)* that partitions users into fine-grained groups based on privatized degrees, drastically shrinking the perturbation domain and amplifying structural signals. Finally, rather than synthesizing a full noise-prone graph, the server performs targeted structural estimation. Extensive experiments on four public datasets demonstrate that SPCC consistently and significantly outperforms state-of-the-art solutions, reducing empirical error across diverse privacy budgets, graph scales, and clique sizes.

CCS Concepts

• **Security and privacy** → **Privacy-preserving protocols**.

Keywords

Local Differential Privacy, Graph Query

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1 Introduction

Cliques [1, 12, 51] are fundamental structures in graphs, capturing the most cohesive relationships among entities. A q-clique is a fully connected subgraph on q nodes [2, 50] where each pair of nodes shares an edge. Clique counting plays a vital role in various analytical tasks, such as measuring clustering coefficients [48], detecting communities [19, 22, 40], predicting missing links [10], identifying functional protein complexes [49], and uncovering collusive behaviors in financial networks [5].

However, releasing clique statistics from real-world networks may introduce substantial privacy risks [8, 30, 34, 50]. Differential Privacy (DP) has increasingly become the *de facto* standard for data privacy and is now widely adopted in graph analysis. Existing DP models are primarily partitioned into two categories: the central model (CDP) [6, 9, 13, 14, 34], which assumes a trusted server to aggregate raw data, and the local model (LDP) [15, 20, 25, 31, 42, 45, 55–57, 59, 60], which eliminates this assumption by having users perturb their data locally before sharing it.

The study of subgraph counting under differential privacy has been extensively explored in both centralized [4, 13, 14, 34] and local [18, 25, 27] models. Accurate clique counting is fundamental to graph analysis, yet remains largely unexplored under LDP. The core challenge is structural fragility: edge-level noise breaks the strict connectivity required by cliques, making them unstable and difficult to quantify.

There are some existing LDP methods customized for triangle counting, a special case of q-clique counting with $q = 3$. They generally follow two technical paradigms.

- **Direct Estimation via Structure Collection.** This paradigm obtains subgraph counts through explicit structure enumeration, which includes methods that reconstruct noisy graphs from perturbed edge information [18, 42, 47], as well as those that extend users' local views for direct counting [25, 26, 38, 50]. While the latter can improve accuracy by assuming multi-hop visibility, both approaches share a fundamental limitation of high communication overhead. Moreover, their reliance on multi-hop visibility assumptions and multi-round interactions limits scalability in large graphs.
- **Indirect Estimation from Local Statistics.** This paradigm collects local statistics from users and employs probabilistic models for subgraph inference [31, 32, 55]. Representative methods like Local2Rounds△ [31] and ARRTwoNS△ [32] improve communication efficiency through edge sampling, but their estimation errors remain highly sensitive to degree, resulting in a high noise sensitivity. The debiasing process in these methods often amplifies variance, particularly in sparse graphs.

However, existing methods following these two paradigms present fundamental limitations. Recent rigorous analyses [18] reveal that fundamental non-interactive edge-level methods inherently suffer from a staggering baseline ℓ_2 error of $O(n^3/\varepsilon^6)$. To bypass this theoretical wall, early two-round protocols like Local2RoundsΔ and ARRTwoNSΔ were proposed, but they still suffer from $O(n^2)$ communication overhead and exhibit ℓ_2 errors of $O(nd_{\max}^3/\varepsilon^2)$. Recent view-collection methods like TDP-VC extend users' local visibility;

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however, they introduce a severe vulnerability to the privacy budget partitioning, resulting in an explosive error bound of $O(nd_{\max}^2/\varepsilon^4)$ in high-privacy regimes. And EdgeOrientΔ leverages graph degeneracy d to achieve $O(nd^2/\varepsilon^4)$ for triangles. Crucially, these methods are over-optimized for $q = 3$ and rely on independent edge perturbation, which inevitably destroys the structural correlations needed for higher-order clique identification ($q > 3$).

To overcome these limitations, we propose a novel LDP framework named SPCC centered on structure preservation. Unlike most existing work focused on triangles (i.e., 3-clique), SPCC can be extended to larger cliques while maintaining high utility. Specifically, SPCC leverages the k-star structure [11, 23], which encapsulates a center node and its k neighbors, as the basic unit for perturbation. Compared with edge-level perturbation, this design preserves inter-edge dependencies and naturally aligns with clique structure. Furthermore, k-stars serve as structural building blocks of dense subgraphs [44], enabling the server to reconstruct cliques directly by validating overlapping k-star reports.

Employing the k-star as the reporting unit presents two main challenges: (i) *degree heterogeneity*, where user degrees vary widely and the length of reported k-star set may reveal private information; and (ii) *selective preservation*, where under a fixed privacy budget, not all neighborhood structures contribute equally to clique estimation accuracy. To address these, we design two mechanisms: (1) *Padding-and-Dropping (PD)* standardizes report lengths to conceal degrees while preserving structure; (2) *Adaptive Grouping Perturbation (AGP)* partitions users by privatized degrees, assigning higher retention probabilities to informative neighbors. Together, *PD* and *AGP* preserve local topology under LDP, achieving the privacy-utility balance.

To summarize, our contributions are as follows.

- We introduce SPCC, the first framework to enable clique counting under a rigorous LDP guarantee.
- SPCC utilizes k-star structure as a perturbation unit, by which it effectively captures graph structure and substantially enhances the accuracy of clique counting.
- SPCC integrates two novel mechanisms, *Padding-and-Dropping (PD)* and *Adaptive Grouping Perturbation (AGP)*, and provides an optimal parameter selection strategy to optimize utility.
- We conducted extensive experiments to validate the effectiveness of SPCC for clique counting. The results demonstrate the superiority of SPCC over state-of-the-art solutions.

The remainder of this paper is organized as follows. Section 2 formalizes the problem and definitions. Section 3 introduces a baseline solution to the clique counting task. Section 4 details our proposed framework SPCC. Section 5 provides theoretical analysis. Section 6 analyzes experimental results. Section 7 reviews related work followed by a conclusion in Section 8.

2 Preliminaries and Problem Statement

This section introduces the preliminaries and the problem statement. We present the problem definition in Section 2.1, the local differential privacy framework in Section 2.2. The Key notations are summarized in Table 1.

Table 1: Summary of notations.

Notation	Definition
$G, \tilde{G}$	Original / noisy graph (V, E)
n, m	#nodes $\lvert V\rvert$, #edges $\lvert E\rvert$
A	Adjacency matrix of G
$N(u)$, d_u	Neighbor set / degree of node u
$d_{\max}$	Max degree
$f^q(G)$, $\hat{f}^q(G)$	True / estimated q-clique count
$\mathcal{KS}_u$	All k-stars centered at u
$\mathcal{K}_u$, $\widetilde{\mathcal{K}}_u$	Original / perturbed k-star set (after *PD*)
κ	A single k-star
q, k	Size of the target clique and local star unit
L	Padding length
$\mathcal{B}, B(v)$	Bucket partition, bucket containing node v
ε, ε_1, ε_2	Total / phase 1 / phase 2 privacy budgets

2.1 Problem Definition

We consider an undirected graph $G = (V, E)$, where V is the node set and $E \subseteq V \times V$ is the edge set. Let $n = |V|$ and $m = |E|$ denote the numbers of nodes and edges, respectively. For a node (i.e., user) $u \in V$, its neighborhood is denoted as $N(u) = \{v \in V \mid (u, v) \in E\}$, and $d_u = |N(u)|$ is the node degree. Alternatively, the neighborhood of a node u can be represented as an adjacency vector $\mathbf{a}_u = (a_{u,1}, \ldots, a_{u,n}) \in \{0, 1\}^n$, where $a_{u,i} = 1$ if the edge $(u, i) \in E$, or otherwise $a_{u,i} = 0$.

In this work, we focus on q-clique, a complete subgraph of G defined over a set of q nodes $S \subseteq V$, i.e., $\forall u, v \in S, (u, v) \in E$. Our objective is to estimate the count of q-cliques in G, denoted as $f^q(G)$. Formally,

$$f^q(G) = |\{S \subseteq V \mid |S| = q \text{ and } \forall u, v \in S, (u, v) \in E\}|. \quad (1)$$

We aim to release an accurate estimate $\hat{f}^q(G)$ of the true count $f^q(G)$ in the context of LDP.

From a node-centric perspective, each node in a q-clique can be viewed as the center of a $(q-1)$-star — a star-shaped substructure where the center node is connected to $q-1$ neighbors. In this work, we propose to use this structure as the unit for data representation and perturbation. Formally, the definition of a k-star is given below.

Definition 1 (k-star). Let $\kappa = (V_\kappa, E_\kappa)$ be an undirected graph. Then κ is called a k-star (or a k-star graph) if

$$V_\kappa = \{u, v_1, v_2, \ldots, v_k\}, \quad (2)$$

$$E_\kappa = \{\{u, v_i\} \mid 1 \le i \le k\}. \quad (3)$$

Notably, in a k-star centered at node u, each neighbor $v_i \in N(u)$ is connected only to the center node u, and there are no edges among neighbors themselves. Formally, the set of all k-stars centered on u is defined as

$$\mathcal{KS}_u = \{\kappa_1, \kappa_2, \ldots, \kappa_t\}, \quad (4)$$

where $t = \binom{d_u}{k}$ is the total number of k-stars centered at u.

2.2 Local Differential Privacy

Local Differential Privacy (LDP) provides a strong, decentralized privacy guarantee by ensuring that data is perturbed at the source

before it leaves a user's device [17, 20, 43, 52], which is formally defined below.

Definition 2 (Local Differential Privacy). A randomized mechanism $\mathcal{M} : \mathcal{X} \to \mathcal{Y}$ satisfies ε-LDP, if for any two inputs $x, x' \in \mathcal{X}$ and any $S \subseteq \mathcal{Y}$:

$$\Pr[\mathcal{M}(x) \in S] \leq e^{\varepsilon} \cdot \Pr[\mathcal{M}(x') \in S]. \tag{5}$$

Here, the privacy budget $\varepsilon > 0$ controls the strength of privacy protection, where a smaller value of ε corresponds to a stronger privacy guarantee.

For graph data, Edge-LDP [24, 35, 37, 45] is a leading privacy model for graph data under LDP. Formally, it is defined over any pair of neighboring adjacency vectors, which differ by at most two bits corresponding to the addition or removal of a single edge.

Definition 3 (Edge-Local Differential Privacy). A local randomizer $\mathcal{R}$ satisfies ε-Edge-LDP, if for any pair of neighboring adjacency vectors $\mathbf{a}_v$ and $\mathbf{a}'_v$ (i.e., differing by the addition or removal of a single edge), and the output sets $S \subseteq \mathcal{Y}$:

$$\Pr[\mathcal{R}(\mathbf{a}_v) \in S] \leq e^{\varepsilon} \cdot \Pr[\mathcal{R}(\mathbf{a}'_v) \in S], \tag{6}$$

To ensure LDP guarantee, Generalized Randomized Response (GRR) [17] and Laplace Mechanism [16] are two classic perturbation mechanisms for categorical and numerical values, respectively.

Generalized Randomized Response (GRR). GRR perturbs an input value x into $\tilde{x}$ following the probability distribution:

$$\Pr[\tilde{x} = y] = \begin{cases} \frac{e^{\varepsilon}}{e^{\varepsilon}+g-1}, & \text{if } y = x \\ \frac{1}{e^{\varepsilon}+g-1}, & \text{otherwise} \end{cases} \tag{7}$$

where g is the domain size. When the input $x \in \{0, 1\}$, i.e., an entry $a_{v,u}$ in the adjacency vector, GRR reduces to Randomized Response (RR) [54] with $g = 2$. By flipping the bit with probability $\frac{1}{e^{\varepsilon}+1}$, this mechanism satisfies ε-Edge-LDP as guaranteed by Definition 3.

Laplace Mechanism. Given a query $f : \mathcal{X} \to \mathbb{R}$, the mechanism injects calibrated noise $\boldsymbol{\eta}$ to the query result $f(x)$,

$$\mathcal{M}_{\text{Lap}}(x) = f(x) + \boldsymbol{\eta}, \tag{8}$$

where the noise $\boldsymbol{\eta} \sim \text{Lap}\left(\frac{\Delta f}{\varepsilon}\right)$ is drawn from a Laplace distribution with scale $\frac{\Delta f}{\varepsilon}$, and $\Delta f = \max_{x,x'} \|f(x) - f(x')\|_1$ is the query's sensitivity, defined over any two neighboring inputs x and x'.

3 EPCC: A Baseline for Clique Counting

We establish the theoretical foundation for q-clique counting under LDP by first introducing EPCC (**E**dge-based **P**rivate **C**lique **C**ounting), grounded in standard unbiased estimation theory [18, 52]. This framework comprises three core stages: *Local Perturbation*, *Algebraic Rescaling*, and *q-Clique Counting*.

Local Perturbation. Each user v independently perturbs their private adjacency vector $\mathbf{a}_v$ into $\tilde{\mathbf{a}}_v$ using Randomized Response (RR), where each bit is retained with probability $p = \frac{e^{\varepsilon}}{e^{\varepsilon}+1}$ and flipped with probability $1 - p = \frac{1}{e^{\varepsilon}+1}$.

Algebraic Rescaling. Directly enumerating cliques on the noisy graph $\tilde{G}$ yields heavily biased results due to the massive number of false-positive edges generated by RR. Instead of physically pruning edges which severely fragments cohesive q-clique structures and leads to severe underestimation, we apply an element-wise algebraic rescaling. For each perturbed edge indicator $X_e \in \{0, 1\}$, the server maps it to a real-valued unbiased estimator Y_e:

$$Y_e = \frac{X_e - (1-p)}{2p - 1} = \frac{X_e \cdot (e^{\varepsilon} + 1) - 1}{e^{\varepsilon} - 1} \tag{9}$$

This linear transformation ensures that the mathematical expectation strictly equals the true state of the edge (i.e., $\mathbb{E}[Y_e] = 1$ if the genuine edge exists, and 0 otherwise).

Algorithm 1: EPCC: Edge-based Private q-Clique Counting Baseline

Input: Graph $G = (V, E)$, privacy budget ε, clique size q
Output: Estimate of the q-clique count $\hat{f}^q(G)$

```
// Local Perturbation
p ← e^ε/(e^ε+1);
foreach u ∈ V do
    ã_u ← RR(a_u, ε) ; // Retain bit w.p. p, flip w.p. 1 − p
    Send ã_u to server;
// Server-side Algebraic Rescaling
foreach potential edge e ∈ (V 2) do
    Y_e ← (X_e − (1−p))/(2p−1) ;   // Compute unbiased edge estimator
// Server-side q-Clique Counting
f̂^q(G) ← 0;
foreach q-node subset S ∈ (V q) do
    f̂^q(G) ← f̂^q(G) + ∏_{e∈(S 2)} Y_e;
return f̂^q(G)
```

q-Clique Counting. Since the localized RR mechanism injects noise independently across all potential edges, the unbiased estimate of a q-clique's presence is simply the product of the unbiased estimators of its constituent edges. The server estimates the global q-clique count, $\hat{f}^q(G)$, by aggregating over all possible q-node subsets $S \in \binom{V}{q}$:

$$\hat{f}^q(G) = \sum_{S \in \binom{V}{q}} \prod_{e \in \binom{S}{2}} Y_e \tag{10}$$

While the EPCC baseline guarantees strict statistical unbiasedness, its application to large-scale graphs is fundamentally hindered by two inherent limitations. First, independent edge perturbation incurs a severe variance explosion. Since the noise accumulates multiplicatively across the global $\binom{n}{q}$ search space, the true structural signals of higher-order cliques are effectively overwhelmed by false positives. Second, evaluating its global unbiased estimator dictates an $O(n^q)$ computational complexity, rendering the algorithm practically intractable for massive graphs. To overcome these theoretical bottlenecks, we propose SPCC in the next section. By elevating the perturbation unit from isolated edges to cohesive k-stars, SPCC effectively restricts the noise domain and transforms the intractable global search into localized vector aggregation, enabling both highly accurate and scalable q-clique counting.

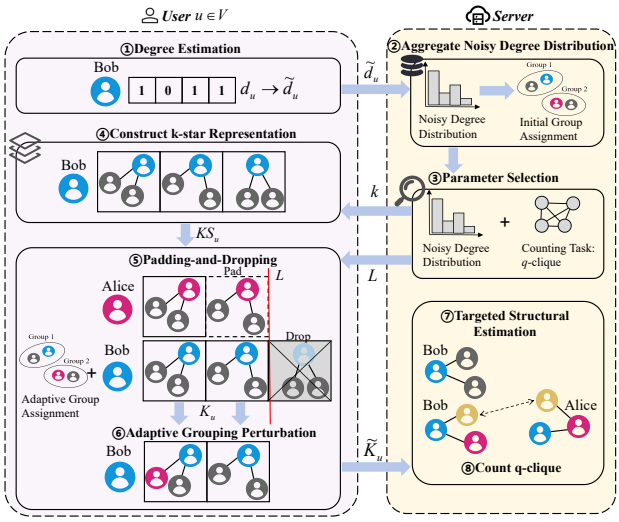


Figure 1: Overview of the proposed SPCC framework.

4 SPCC: An Optimized Framework for Clique Counting via k-Stars

4.1 Overview

To address the limitations of the baseline method, we propose SPCC (**S**tar-based **P**rivate **C**lique **C**ounting), for q-clique counting with enhanced utility and efficiency. SPCC represents a paradigm shift in LDP graph analysis by redefining the fundamental perturbation unit from isolated edges to cohesive k-stars, which effectively encapsulate local topological context.

Algorithm 2: SPCC: Star-based Private q-Clique Counting Framework

Input: Graph $G = (V, E)$, budget $\varepsilon = \varepsilon_1 + \varepsilon_2$, clique size q
Output: Estimate of the q-clique count $\hat{f}^q(G)$

```
// Phase 1: Degree Estimation & Parameter Setup
```
1 **foreach** *user* $u \in V$ **do**
2 $\quad \tilde{d}_u \leftarrow d_u + \mathrm{Lap}(2/\varepsilon_1)$;
3 $\quad$ Send $\tilde{d}_u$ to server;

```
// Server Parameter Selection
```
4 $\hat{d}_{\max} \leftarrow \max_{u \in V}\{\tilde{d}_u\}$;
5 $k \leftarrow \mathrm{ComputeOptimalK}(q)$; // Via Eq. 20
6 $L \leftarrow \mathrm{ComputeOptimalL}(\{\tilde{d}_u\}, k, \varepsilon_2)$; // Via Eq. 22
7 $\mathcal{B} \leftarrow \mathrm{Partition}(V, \{\tilde{d}_u\}, L)$; // Via Eq. 11 and 12
8 Broadcast $(\hat{d}_{\max}, \mathcal{B}, k, L)$ to all users;

```
// Phase 2: Local Perturbation & Data Collection
```
9 **foreach** *user* $u \in V$ **do**
10 $\quad \widetilde{\mathcal{K}}_u \leftarrow \mathrm{LocalRandomizer}(N(u), \hat{d}_{\max}, \mathcal{B}, k, L, \varepsilon_2)$; // Call Alg. 3
11 $\quad$ Send $\widetilde{\mathcal{K}}_u$ to server;
12 $\hat{f}^q(G) \leftarrow \mathrm{TargetedStructuralEstimation}(\{\widetilde{\mathcal{K}}_u\}_{u \in V}, \mathcal{B}, k, L, \varepsilon_2)$; // Call Alg. 4
13 **return** $\hat{f}^q(G)$

As shown in Fig. 1, the SPCC framework comprises two phases. Phase 1 estimates the degree distribution to establish a privatized view of the whole graph and establishes optimized global parameters, which Phase 2 then utilizes to collect and perturb local structural information from users; the total privacy budget ε is partitioned into ε_1 for degree perturbation and ε_2 for k-star perturbation, satisfying $\varepsilon_1 + \varepsilon_2 = \varepsilon$.

Phase 1. Each user u first perturbs their true degree d_u with Laplace noise under ε_1 and reports the noisy degree $\tilde{d}_u$ to the server. Based on the aggregated noisy degree distribution, the server identifies the maximum noisy degree $\hat{d}_{\max}$[1], and computes the theoretically optimal k-star size k and padding length L. Subsequently, leveraging L and the noisy distribution, the server partitions users into adaptive degree-homogeneous buckets $\mathcal{B}$. Finally, the server broadcasts these global parameters $(\hat{d}_{\max}, \mathcal{B}, k, L)$ to all users.

Phase 2. Upon receiving the global parameters, each user u first performs a graph projection step, uniformly sampling to retain at most $\hat{d}_{\max}$ neighbors. From neighborhood $\widehat{N}(u)$, user u constructs a local set of k-stars $\mathcal{KS}_u$, applies the *PD* mechanism to standardize the report length to L, and executes *AGP* for localized perturbation within their bucket. Finally, the server aggregates these perturbed k-stars into intermediate localized neighborhoods, performing efficient targeted estimations to compute $f^q(G)$.

In what follows, we elaborate on the implementation details of SPCC. Algorithm 2 shows the workflow.

4.2 Degree Estimation

Phase 1 encompasses degree perturbation, global parameter optimization, and degree-aware grouping based on the noisy degrees. In this subsection, we focus on the degree estimation component, while the grouping mechanism will be detailed in Section 4.4 as part of the Adaptive Grouping Perturbation (*AGP*) analysis.

The group-wise perturbation in Phase 2 requires user-degree information, but using true degrees risks privacy leakage. To address this, each user u first perturbs their true degree d_u with Laplace noise (Line 2 of Algorithm 2): $\tilde{d}_u = d_u + \mathrm{Lap}(2/\varepsilon_1)$ (the addition or removal of an undirected edge affects the degrees of two incident nodes). The result is rounded to the nearest integer for practicality and outlier control. Users then send $\tilde{d}_u$ to the server, which aggregates them into a noisy degree distribution. Then the server identifies the maximum noisy degree $\hat{d}_{\max} \leftarrow \max_{u \in V}\{\tilde{d}_u\}$, which serves as the threshold for the graph projection (Line 4 of Algorithm 2).

With the noisy degree distribution collected, the server handles the global parameter setup (optimal k and L) and the grouping assignment $\mathcal{B}$. The detailed mechanisms for parameter selection and grouping are deferred to Section 4.6 and Section 4.4, respectively.

4.3 k-star: A Local Structure Representation

Leveraging the local neighborhood $\widehat{N}(u)$, each user constructs a local data representation, which is in the form of a set of k-stars (Line 2 of Algorithm 3). Upon receiving optimal k from the server,

[1]Following standard practice in LDP graph analysis, both the baseline methods and our proposed framework employ a graph projection step to strictly bound the maximum node degree by a threshold $\hat{d}_{\max}$.

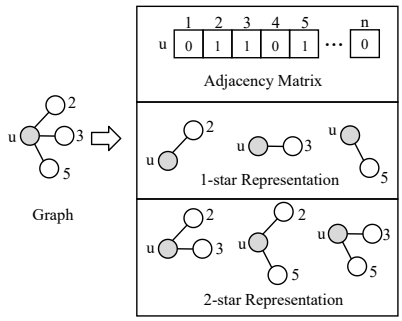


Figure 2: A case for k-star representation.

user u enumerates $\binom{d_u}{k}$ k-stars centered at u. By organizing multiple edges into cohesive structural units, the k-star representation preserves essential local structure.

Example. We take node u in Fig. 2 (with neighbors $N(u) = \{2, 3, 5\}$) as an example. When $k = 1$, the k-star set is $\mathcal{KS}_u = \{\{2\}, \{3\}, \{5\}\}$; when $k = 2$, the k-star set is $\mathcal{KS}_u = \{\{2, 3\}, \{2, 5\}, \{3, 5\}\}$.

The k-star representation is designed to achieve three key objectives within the SPCC framework:

- **Improving preservation opportunities.** Each edge (u, v) is inherently embedded in $\binom{d_u-1}{k-1}$ distinct k-stars centered at u. This structural redundancy ensures that even under a limited reporting budget (where only a subset of k-stars is sampled), an edge still has multiple opportunities to be preserved.
- **Improving noise resilience through structural consistency.** While independent edge reporting allows spurious edges to form false cliques, a k-star requires joint consistency of all k neighbors. The probability that noise simultaneously creates a consistent k-star is $O(p_{\text{err}}^k)$, where p_{err} is the error probability per-node. For $k \geq 2$, this makes false k-stars exponentially rarer than false edges, significantly suppressing noise in clique counting.
- **Improving information efficiency.** Reporting k-stars requires only $O(k \log n)$ bits per star—the same asymptotic cost as reporting k independent edges. However, k-stars encode additional structural semantics: the server knows the reported neighbors form a *complete star* with the center, directly serving as building blocks for q-cliques. This structural awareness reduces combinatorial search during reconstruction.

4.4 Local Randomizer

For each node u, the neighborhood information has been represented as a set of k-stars centered at node u. Based on this representation, we proceed to the local randomizer, which addresses bandwidth efficiency and degree privacy. Specifically, we design two synergistic mechanisms: *Padding-and-Dropping (PD)* which enforces structural uniformity, and *Adaptive Grouping Perturbation (AGP)*, which optimizes the perturbation domain via topological priors.

4.4.1 Padding-and-Dropping Mechanism. In this mechanism, all users standardize their k-star sets to a universal length L, determined by the server in Phase 2 via the parameter selection module in Section 4.6. Each user u processes their local set $\mathcal{KS}_u$ as follows (Line 3 of Algorithm 3):

- **Padding:** If $|\mathcal{KS}_u| < L$, the set is padded by $L - |\mathcal{KS}_u|$ *dummy k-stars*. A dummy k-star consists of k neighbor IDs all set to a special **dummy identifier** $\text{dum} = n + 1$, where n is the total number of nodes. This identifier is strictly outside the valid node ID range $\{1, \ldots, n\}$.
- **Dropping:** If $|\mathcal{KS}_u| \geq L$, $|\mathcal{KS}_u| - L$ genuine k-stars are randomly dropped from the set.

Consequently, the *PD* mechanism ensures that any specific genuine k-star $\kappa \in \mathcal{KS}_u$ is included in the reported set $\mathcal{K}_u$ with a probability $\rho_u = \min(1, L/|\mathcal{KS}_u|)$. This procedure ensures that each user reports a set of k-stars with a standardized length L (Line 10 of Algorithm 2). The choice of L balances a theoretical trade-off: a small L drops genuine k-stars (losing structural information), while a large L pads many dummies (increasing overall perturbation variance).

4.4.2 Adaptive Grouping Perturbation (AGP). After applying the *PD* mechanism, each user holds a standardized set of k-stars $\mathcal{K}_u$. A conventional application of Generalized Randomized Response (GRR) [17] over the global vertex domain $\mathcal{V}$ would result in a prohibitively low retention probability $p = e^{\varepsilon'}/(e^{\varepsilon'} + n - 1)$ (where $\varepsilon' = \varepsilon_2/(kL)$), rendering the recovery of higher-order structures practically infeasible due to excessive noise injection.

Alignment with Topological Priors. Real-world networks inherently exhibit severe structural heterogeneity, typically characterized by power-law degree distributions and heavy-tail hubs. Under LDP, these high-degree hubs predominantly dictate the presence of higher-order cliques, but they simultaneously act as the primary catalyst for local variance explosion if subjected to global perturbation. To rigorously amplify structural signals without inflating the privacy budget, *AGP* leverages this topological prior. Departing from rigid grouping, we design an adaptive partitioning strategy that dynamically segments the vertex set V into disjoint, degree-homogeneous buckets $\mathcal{B} = \{B_j\}_{j=1}^{|\mathcal{B}|}$. This partition is derived from the noisy degree distribution $\{\tilde{d}_v\}_{v \in V}$ generated from Phase 1, ensuring zero additional privacy leakage.

Formally, for each bucket $B_j \in \mathcal{B}$, let $\mu(B_j)$ and $\sigma(B_j)$ denote the mean and standard deviation of the noisy degrees within B_j. The partition $\mathcal{B}$ is generated by sorting V and segmenting it under two constraints: size constraint (Eq. 11) and purity constraint (Eq. 12).

$$|B_j| \leq C(\mu(B_j)) = \begin{cases} \lceil \alpha L \rceil, & \text{if } \mu(B_j) > q_{90} \\ \lceil \lambda L \rceil, & \text{if } q_{50} < \mu(B_j) \leq q_{90} \\ B_{\max} = \lceil \beta L \rceil, & \text{if } \mu(B_j) \leq q_{50} \end{cases} \tag{11}$$

$$\sigma(B_j) \leq \eta \cdot \mu(B_j) \tag{12}$$

The size constraint systematically segments the topology using structural percentile thresholds (q_{50}, q_{90}) and size scaling factors $(\alpha < 1 \leq \lambda < \beta)$ to implement a degree-aware privacy budget allocation. Specifically, q_{90} essentially *quarantines* the top 10% of heavy-tail hubs into tightly constrained local domains ($\alpha < 1$). Since the localized GRR variance scales linearly with the bucket size, this restriction elevates the retention probability p for these

critical structures. Conversely, q_{50} aggregates the prevalent *long-tail* of low-degree nodes into coarser buckets ($\beta > 1$), effectively mitigating background noise through dilution. These scaling factors function as dynamic regularizers rather than fixed constants: $\alpha \propto (1-Gini)$ [7] selectively shrinks the perturbation domain for highly skewed graphs to safeguard hubs, while $\beta \propto (1 + e^{-\varepsilon_2})$ expands under stringent privacy regimes to absorb padding noise.

Complementing this, the purity constraint ensures strict intra-bucket homogeneity against the initial noise injected during degree estimation. We introduce η as a variance tolerance parameter, typically set to 0.2, to reconcile the trade-off between fine-grained clustering and sufficient domain sizes. Any intermediate bucket violating this purity requirement is recursively split at its median. This recursive refinement guarantees that nodes within a shared perturbation domain inherently share highly similar topological characteristics, which ultimately determines the final number of buckets $|\mathcal{B}|$.

Intra-Bucket Local Perturbation. *AGP* significantly amplifies structural signals by confining the (noisy) perturbation domain. When perturbing a k-star $\kappa = v_1, \ldots, v_k$ of user u, each neighbor v_i is not perturbed over the global vertex set, but rather over the restricted domain $B^+(v_i) = B(v_i) \cup \text{dum}$, where $B(v_i)$ is its adaptively assigned bucket from Phase 1. This domain reduction directly boosts the retention probability.

To strictly maintain degree concealment, users may need to pad their reports with dummy items. To ensure these dummies do not skew the localized variance or break indistinguishability, they must adhere to the exact same localized mechanism. To achieve this, whenever a dummy is required, the user uniformly samples a random node $x \in V$ from the global vertex set. This intrinsically maps the dummy to a target bucket $B_j \in \mathcal{B}$ with probability $|B_j|/n$, ensuring padding noise is proportionally and uniformly distributed across the graph structure.

Let x represent either a genuine target v_i or a sampled dummy target. The input x is perturbed into an output $y \in B^+(x)$ following a unified localized GRR distribution:

$$\Pr[\tilde{v}_i = y \mid x, \mathcal{B}] = \begin{cases} \frac{e^{\varepsilon'}}{e^{\varepsilon'}+|B^+(x)|-1}, & \text{if } y = x \\ \frac{1}{e^{\varepsilon'}+|B^+(x)|-1}, & \text{if } y \in B^+(x) \setminus \{x\} \end{cases} \tag{13}$$

This unified formulation renders genuine edges and dummy padding computationally indistinguishable post-perturbation. Furthermore, because the bucket partition $\mathcal{B}$ is deterministically generated from the ε_1-LDP degree distribution (Phase 1), it constitutes public knowledge according to the post-processing property of differential privacy. Thus, this domain reduction inherently avoids privacy leakage and rigorously satisfies ε-Edge-LDP (formalized in Section 5.1).

Amplification of Joint Structural Signals. By effectively focusing the probability mass within degree-homogeneous buckets, this mechanism significantly improves the retention probability of any node when it appears as a neighbor in a k-star. Specifically, for an edge (u, v), node v's individual retention probability is $p_v = \frac{e^{\varepsilon'}}{e^{\varepsilon'}+|B^+(v)|-1}$, which depends only on v's own bucket $B^+(v)$. Since the domain size $|B^+(v)|$ is tightly bounded by the *AGP* mechanism and satisfies $|B^+(v)| \ll n$, this localized retention rate p_v is orders of magnitude larger than that of the naive global GRR. Combined with the inherent redundancy of the k-star representation—where each true edge (u, v) appears in $R_{uv} = \binom{d_u-1}{k-1}$ distinct k-stars—the overall probability that an edge survives the perturbation becomes $1 - (1 - p_v)^{R_{uv}}$, which remains highly robust for targeted structural estimation.

By effectively focusing the probability mass within degree-homogeneous buckets, this mechanism significantly improves the retention probability of any node when it appears as a neighbor in a k-star. Specifically, for an edge (u, v), node v's individual retention probability is $p_v = \frac{e^{\varepsilon'}}{e^{\varepsilon'}+|B^+(v)|-1}$, which depends only on v's own bucket $B^+(v)$.

Algorithm 3: Local Randomizer (User)

Input: Neighborhood $N(u)$, threshold $\hat{d}_{\max}$, grouping $\mathcal{B}$, k, L, budget ε_2
Output: Perturbed k-star set $\widetilde{\mathcal{K}}_u$

```
// Graph Projection & k-star Construction
```
$\widehat{N}(u) \leftarrow \text{GraphProjection}(N(u), \hat{d}_{\max})$;
$\mathcal{KS}_u \leftarrow \{\text{all } k\text{-subsets in } \widehat{N}(u)\}$;
```
// Padding-and-Dropping (PD)
```
$\mathcal{K}_u \leftarrow PD(\mathcal{KS}_u, L)$; // Standardize length to L
```
// Adaptive Grouping Perturbation (AGP)
$\varepsilon' \leftarrow \varepsilon_2/(k \cdot L)$;
**foreach** $\kappa \in \mathcal{K}_u$ **do**
  **foreach** $v_i \in \kappa$ **do**
    $\tilde{v}_i \leftarrow \begin{cases} v_i & \text{w.p. } \frac{e^{\varepsilon'}}{e^{\varepsilon'}+|B^+(v_i)|-1} \\ y \in B^+(v_i) \setminus \{v_i\} & \text{w.p. } \frac{1}{e^{\varepsilon'}+|B^+(v_i)|-1} \end{cases}$;
    $\tilde{\kappa} \leftarrow \tilde{\kappa} \cup \{\tilde{v}_i\}$;
  $\widetilde{\mathcal{K}}_u \leftarrow \widetilde{\mathcal{K}}_u \cup \{\tilde{\kappa}\}$;
**return** $\widetilde{\mathcal{K}}_u$

## 4.5 Targeted Structural Estimation

**Algorithm 4:** Targeted Structural Estimation (Server)

**Input:** Perturbed reports $\{\widetilde{\mathcal{K}}_u\}_{u \in V}$, grouping $\mathcal{B}$, $k$, $L$, budget $\varepsilon_2$
**Output:** Estimate of $q$-clique count $\hat{f}^q(G)$

$\varepsilon' \leftarrow \varepsilon_2/(k \cdot L)$;
**foreach** *user* $u \in V$ **do**
  $\mathcal{V}_u \leftarrow \{v \mid v \in \tilde{\kappa}, \tilde{\kappa} \in \widetilde{\mathcal{K}}_u\}$
  **foreach** *target node* $v \in \mathcal{V}_u$ **do**
    $p_v \leftarrow \frac{e^{\varepsilon'}}{e^{\varepsilon'}+|B^+(v)|-1}$ ; // AGP retention probability
    $\hat{R}_{uv} \leftarrow \binom{\hat{d}_u-1}{k-1}$ ; // Estimated structural redundancy
    // Algebraic rescaling for observed structural supports
    $\hat{A}_{u,v} \leftarrow \frac{1}{\hat{R}_{uv}} \sum_{\tilde{\kappa} \in \widetilde{\mathcal{K}}_u} \sum_{\tilde{v} \in \tilde{\kappa}} \frac{\mathbf{1}_{\{\tilde{v}=v\}} - \frac{1-p_v}{|B^+(v)|-1}}{p_v - \frac{1-p_v}{|B^+(v)|-1}}$;
$\mathcal{S}_{cand} \leftarrow \bigcup_{u \in V} \{S \subseteq V : |S| = q, S \text{ is supported by } \widetilde{\mathcal{K}}_u\}$
$\hat{f}^q(G) \leftarrow \sum_{S \in \mathcal{S}_{cand}} \prod_{(u,v) \in \binom{S}{2}} \hat{A}_{u,v}$;
**return** $\hat{f}^q(G)$

To reconstruct the target graph structures from the perturbed reports, SPCC employs a two-stage Targeted Structural Aggregation.
```

De-biasing *AGP*. Recall from Section 4.4.2 that when a user u perturbs a k-star $\kappa = \{v_1, \ldots, v_k\}$, each original neighbor v_i is independently perturbed into $\tilde{v}_i$ over its bucket $B^+(v_i)$ according to Eq. 13. To debias the estimation, for any observed perturbed node $\tilde{v}$ (which corresponds to $\tilde{v}_i$ in the algorithm) and target node v, we define an algebraically rescaled indicator variable $Y_{\tilde{v}}(v)$:

$$Y_{\tilde{v}}(v) = \frac{\mathbf{1}_{\{\tilde{v}=v\}} - \frac{1-p_v}{|B^+(v)|-1}}{p_v - \frac{1-p_v}{|B^+(v)|-1}} \tag{14}$$

where $p_v = \frac{e^{\varepsilon'}}{e^{\varepsilon'}+|B^+(v)|-1}$ is the localized retention probability for node v. The term $\frac{1-p_v}{|B^+(v)|-1}$ represents the uniform flip probability over the remaining perturbation domain. This linear transformation dynamically adjusts to the *AGP* partition, ensuring that $\mathbb{E}[Y_{\tilde{v}}(v)]$ strictly equals 1 if the true node was indeed v, and 0 otherwise.

Structural Aggregation. To estimate the presence of an edge (u, v), the server aggregates the rescaled $Y_{\tilde{v}}(v)$ across the L reported k-stars in $\widetilde{\mathcal{K}}_u$.

During the aggregation process, the server remains agnostic to the exact local pool size $|\mathcal{KS}_u|$. The server approximates these structural parameters using the Phase 1 privatized degree $\tilde{d}_u$ (Line 6 of Algorithm 4). The expected structural redundancy is thus estimated as $\hat{R}_{uv} = \binom{\tilde{d}_u - 1}{k-1}$. The server formulates the edge estimator $\hat{A}_{u,v}$ (Line 7 of Algorithm 4):

$$\hat{A}_{u,v} = \frac{1}{\hat{R}_{uv}} \sum_{\tilde{\kappa} \in \widetilde{\mathcal{K}}_u} \sum_{\tilde{v} \in \tilde{\kappa}} Y_{\tilde{v}}(v) \tag{15}$$

Conditioned on the public outputs of Phase 1, $\hat{R}_{uv}$ acts as a deterministic constant. The algebraic rescaling in Eq. 15 is specifically designed to de-bias the GRR noise introduced by the *AGP* mechanism. Regarding the controlled downward bias introduced by *PD*, rather than locally amplifying the perturbed estimates to offset this loss—which would trigger severe variance explosion—SPCC systematically addresses this underestimation at the system level. The trade-off between this structural dropping bias and the GRR perturbation variance is mathematically resolved by our optimal padding length L^* (Section 4.6.2), which minimizes the total ℓ_2-error.

q-Clique Counting. Finally, the server aggregates $\hat{A}_{u,v}$ to compute the global count $\hat{f}^q(G)$. SPCC restricts the estimation to a candidate set $\mathcal{S}_{cand}$ (Line 8 of Algorithm 4), defined as the union of all potential q-cliques identified from the individual structural reports:

$$\mathcal{S}_{cand} = \left\{ S \in \binom{V}{q} : \exists u \in S, \exists \tilde{\kappa} \in \widetilde{\mathcal{K}}_u \text{ s.t. } S \subseteq (\tilde{\kappa} \cup \{u\}) \right\} \tag{16}$$

$$\hat{f}^q(G) = \sum_{S \in \mathcal{S}_{cand}} \prod_{(u,v) \in \binom{S}{2}} \hat{A}_{u,v} \tag{17}$$

This candidate-driven discovery guarantees that a q-clique remains in the estimation domain provided that at least one constituent node reports a supporting k-star structure, thereby maximizing the structural recall.

4.6 Parameter Selection

In Phase 2, the server selects key parameters based on the q-clique task and privatized degree distribution: the k-star size k and padding length L (Algorithm 2, lines 6–7). These parameters govern the structural granularity and noise characteristics of the local randomizer.

4.6.1 Optimal Choice of k-star size. The selection of k governs the fundamental trade-off between structural fidelity and perturbation noise.

For an edge $(u, v) \in E$, recall from Section 4.4 that each neighbor v in a k-star of user u is perturbed using GRR with retention probability $p_v = \frac{e^{\varepsilon'}}{e^{\varepsilon'}+|B^+(v)|-1}$. This edge appears in $\binom{d_u-1}{k-1}$ distinct k-stars centered at u. The probability that edge (u, v) is preserved in at least one k-star report is:

$$P_{\text{retain}}(k) = 1 - (1 - p_v)^{\binom{d_u-1}{k-1}}. \tag{18}$$

We define the utility function for q-clique counting as the expected retention rate of clique-forming edges:

$$\mathcal{U}(k) = \mathbb{E}\left[P_{\text{retain}}(k)\right], \tag{19}$$

where the expectation is taken over the degree distribution and bucket assignments.

Since $\binom{d_u-1}{k-1}$ is monotonically increasing with k for $k \le d_u$, the utility $\mathcal{U}(k)$ is maximized when k takes the largest feasible value. However, for $k > q-1$, the k-star enumeration introduces extraneous edges outside any q-clique, adding noise without structural benefit. For $k < q-1$, the representation fails to fully capture the clique's connectivity pattern.

Therefore, an optimal k, which ensures complete capture of q-clique structures and maximize edge retention probability under privacy constraints, is:

$$k^* = \arg\max_{k \in [1, q-1]} \mathcal{U}(k) = q - 1. \tag{20}$$

4.6.2 Optimal Choice of Padding Length. The choice of L presents a critical trade-off: a small L causes excessive dropping (high bias), while a large L introduces many dummies, increasing perturbation variance. We need to formalize this trade-off and derive the optimal.

Bias. When $L < |\mathcal{KS}_u|$, random dropping of genuine k-stars induces downward bias (underestimation), with an expected loss of $\max(0, |\mathcal{KS}_u| - L)$ stars per user. Let c_s denote the expected marginal contribution of a single k-star to the global q-clique count, the bias introduced by user u is approximately:

$$\text{Bias}_{\text{drop}}(u) = c_s \cdot \max(0, |\mathcal{KS}_u| - L). \tag{21}$$

When $L > |\mathcal{KS}_u|$, dummy k-stars (ID dum) are added. While the GRR noise implies that a dummy item may randomly flip into a genuine vertex ID (physically creating a false edge), our algebraic rescaling step (Eq. 15) mathematically neutralizes this effect. Consequently, padding intrinsically avoids inflating the expected clique counts; it exclusively contributes to the perturbation variance without injecting bias.

Variance. After *PD*, each user reports exactly L k-stars (some genuine, some dummy). Each neighbor ID within these stars is perturbed via GRR within its assigned bucket. The variance contributed by each user scales approximately as $O(L \cdot \frac{1}{\varepsilon'+\sqrt{n}})$. For typical parameter regimes where ε' is small relative to $\sqrt{n}$, this simplifies to $\text{Var}(L) \approx \gamma \cdot L$ for constant $\gamma > 0$.

Combining our bias and variance analyses, the theoretical ℓ_2-error can be approximated and minimized:

$$L^* = \arg\min_L \left[\underbrace{\left(c_s \cdot \sum_{u \in V} \max(0, |\mathcal{KS}_u| - L) \right)^2}_{\text{dropping bias}} + \underbrace{\gamma \cdot n \cdot L}_{\text{padding noise}} \right], \quad (22)$$

In practice, the exact local sizes $|\mathcal{KS}_u|$ in Eq. 22 are unknown to the server. Therefore, the server approximates this theoretical objective and computes L^* efficiently by replacing $|\mathcal{KS}_u|$ with their privatized structural estimates $\binom{\tilde{d}_u}{k}$, utilizing the noisy degree distribution obtained in Phase 1. Once L^* is determined, the server sets the global padding parameter $L = L^*$ to strictly regulate the perturbation variance across all users in Phase 2.

The parameters γ and c_s are estimated directly from privatized data: γ is analytically derived from the Phase 2 privacy budget ε_2 and the localized GRR parameters completely determined by the Phase 1 noisy degrees. While c_s, representing the expected number of q-cliques sharing a given k-star, is estimated via the privatized edge density $p_{\text{edge}} \approx \frac{1}{n(n-1)} \sum_{u \in V} \tilde{d}_u$. By approximating the underlying graph with this uniform density, the expected contribution is $c \approx \binom{n-k-1}{q-k-1} \cdot p_{\text{edge}}^{\binom{q}{2}-k}$.

5 Theoretical Analysis

5.1 ε-Edge-LDP Guarantee

Theorem 5.1. The complete framework satisfies ε-Edge-LDP.

Proof. Consider any two adjacent graphs G and G' that differ by a single edge (u, v). This discrepancy exclusively impacts the local star-set $\mathcal{KS}_u$, which is subsequently standardized by the *PD* mechanism into a set $\mathcal{K}_u$ of fixed length containing exactly $k \cdot L$ identifiers.

The privacy guarantee of our mechanism, denoted as $\mathcal{M}$, is determined by the sequential composition of two randomized phases: $\mathcal{M}_1$, which generates the adaptive bucket partition $\mathcal{B}$ under ε_1-LDP, and $\mathcal{M}_2$, which applies localized GRR to the $k \cdot L$ identifiers. For any generated output sequence $\widetilde{\mathcal{K}}_u$, the marginal probability over the entire partition space Ω is defined as:

$$\Pr[\mathcal{M}(G) = \widetilde{\mathcal{K}}_u] = \sum_{\mathcal{B} \in \Omega} \Pr[\mathcal{M}_1(G) = \mathcal{B}] \cdot \Pr[\mathcal{M}_2(\mathcal{K}_u \mid \mathcal{B}) = \widetilde{\mathcal{K}}_u] \quad (23)$$

By the strict ε_1-LDP guarantee of Phase 1, the partition generation satisfies $\Pr[\mathcal{M}_1(G) = \mathcal{B}] \le e^{\varepsilon_1} \Pr[\mathcal{M}_1(G') = \mathcal{B}]$ for all valid partitions $\mathcal{B} \in \Omega$. Substituting this inequality into Eq. (23) yields:

$$\Pr[\mathcal{M}(G) = \widetilde{\mathcal{K}}_u] \le e^{\varepsilon_1} \sum_{\mathcal{B} \in \Omega} \Pr[\mathcal{M}_1(G') = \mathcal{B}] \cdot \Pr[\mathcal{M}_2(\mathcal{K}_u \mid \mathcal{B}) = \widetilde{\mathcal{K}}_u] \quad (24)$$

In Phase 2, each identifier $v_i \in \mathcal{K}_u$ is independently perturbed into $\tilde{v}_i \in B^+(v_i)$ with budget of $\varepsilon' = \varepsilon_2/(k \cdot L)$. Since $\mathcal{M}_1$ relies on a noise mechanism with an infinite support set, $\mathcal{B}$ acts as a random variable. Marginalized over this randomized partition space $\mathcal{B} \sim \mathcal{M}_1(G')$, the localized GRR distribution (defined in Eq. 7) bounds the maximum likelihood ratio for any pairwise substitution $v_i \to v_i'$:

$$\sum_{\mathcal{B} \in \Omega} \Pr[\mathcal{M}_1(G') = \mathcal{B}] \cdot \frac{\Pr[\tilde{v}_i = y \mid v_i, \mathcal{B}]}{\Pr[\tilde{v}_i = y \mid v_i', \mathcal{B}]} \le e^{\varepsilon'} \quad (25)$$

Due to the independence of the perturbations across the standardized sequence, the cumulative privacy loss for $\mathcal{K}_u$ is strictly bounded by the product of the individual bounds. Thus, over the expected partition space, we have:

$$\begin{aligned} &\sum_{\mathcal{B} \in \Omega} \Pr[\mathcal{M}_1(G') = \mathcal{B}] \cdot \Pr[\mathcal{M}_2(\mathcal{K}_u \mid \mathcal{B}) = \widetilde{\mathcal{K}}_u] \\ &\le e^{k \cdot L \cdot \varepsilon'} \sum_{\mathcal{B} \in \Omega} \Pr[\mathcal{M}_1(G') = \mathcal{B}] \cdot \Pr[\mathcal{M}_2(\mathcal{K}_u' \mid \mathcal{B}) = \widetilde{\mathcal{K}}_u] \end{aligned} \quad (26)$$

Given the privacy budget allocation $\varepsilon' = \varepsilon_2/(k \cdot L)$, the cumulative bound simplifies to $e^{k \cdot L \cdot \varepsilon'} = e^{\varepsilon_2}$. Substituting Eq. (26) back into Eq. (24), we derive the final formulation:

$$\begin{aligned} \Pr[\mathcal{M}(G) = \widetilde{\mathcal{K}}_u] &\le e^{\varepsilon_1} \cdot e^{\varepsilon_2} \sum_{\mathcal{B} \in \Omega} \Pr[\mathcal{M}_1(G') = \mathcal{B}] \cdot \Pr[\mathcal{M}_2(\mathcal{K}_u' \mid \mathcal{B}) = \widetilde{\mathcal{K}}_u] \\ &= e^{\varepsilon_1 + \varepsilon_2} \Pr[\mathcal{M}(G') = \widetilde{\mathcal{K}}_u] \end{aligned} \quad (27)$$

Therefore, the joint framework strictly satisfies pure ε-Edge-LDP, where $\varepsilon = \varepsilon_1 + \varepsilon_2$. □

5.2 Analysis of the total theoretical ℓ_2-Error

Theorem 5.2. For $q \ge 3$, let $\varepsilon = \varepsilon_1 + \varepsilon_2$ be the total privacy budget partitioned into Phase 1 (ε_1) and Phase 2 (ε_2). Given the optimal k-star size $k^* = q - 1$ and padding length L^*, the total estimation error $\mathbb{E}[(\hat{f}^q(G) - f^q(G))^2]$ is bounded by:

$$O\left(\frac{n}{\varepsilon_1^2} + \frac{n B_{\max} \hat{d}_{\max}^{q-1} L^{*2}}{\varepsilon_2^2} + \min\left(n \hat{d}_{\max}^{q-1} e^{-c \hat{d}_{\max}^{q-2}}, \frac{n^2 \hat{d}_{\max}^{2q-4}}{L^*} \right) \right). \quad (28)$$

Proof. The total ℓ_2-error decomposes into three independent components: degree perturbation variance (Var_{deg}), GRR perturbation variance (Var_{GRR}), and *PD* dropping bias ($\text{Bias}^2_{\text{drop}}$).

1. Variance of Phase 1 (Degree Perturbation). Each node u adds Laplace noise to its degree. Since nodes are independent, the aggregated variance is:

$$\text{Var}_{\text{deg}} = \sum_{u \in V} \text{Var}(\text{Lap}(2/\varepsilon_1)) = O(n/\varepsilon_1^2). \quad (29)$$

2. Variance of Phase 2 (GRR Perturbation). Let $Y_{\tilde{v}}(v)$ be the unbiased estimator for a retained identifier v under *AGP* where the perturbation domain size is bounded by the maximum bucket capacity $B_{\max}$. Given $\varepsilon' = \varepsilon_2/(k^* L^*)$, by Taylor expansion of the GRR variance formula:

$$\text{Var}(Y_{\tilde{v}}(v)) = \frac{e^{\varepsilon'} + B_{\max} - 2}{(e^{\varepsilon'} - 1)^2} \approx \frac{B_{\max}}{\varepsilon'^2} = O\left(\frac{B_{\max} k^{*2} L^{*2}}{\varepsilon_2^2} \right). \quad (30)$$

For an edge (u, v), the expected number of retained observations is $M_{uv} = R_{uv} \rho_u$, where $R_{uv} = \binom{d_u - 1}{q-2}$ is the structural redundancy

and $\rho_u = \min(1, L^*/\binom{d_u}{k^*})$ is the *PD* retention rate. Since the estimator is formulated as $\hat{A}_{u,v} = \frac{1}{R_{uv}} \sum Y_{\tilde{v}}(v)$, its variance is:

$$\mathrm{Var}(\hat{A}_{u,v}) = \frac{M_{uv}}{R_{uv}^2}\mathrm{Var}(Y_{\tilde{v}}(v)) = \frac{\rho_u}{R_{uv}}\mathrm{Var}(Y_{\tilde{v}}(v)) = O\left(\frac{\rho_u B_{\max} L^{*2}}{R_{uv}\varepsilon_2^2}\right). \tag{31}$$

Summing over $|E| = O(n\hat{d}_{\max})$ potential edges with structural sensitivity $C_e = O(\hat{d}_{\max}^{q-2})$. The accumulated GRR variance is strictly bounded by:

$$\begin{aligned}\mathrm{Var}_{\mathrm{GRR}} &= \sum_{e\in E} C_e^2 \mathrm{Var}(\hat{A}_{u,v}) \le O\left(\sum_{e\in E}\frac{C_e^2\rho_u}{R_{uv}}\frac{B_{\max}L^{*2}}{\varepsilon_2^2}\right)\\ &\le O\left(n\hat{d}_{\max}\cdot\hat{d}_{\max}^{q-2}\cdot\frac{B_{\max}L^{*2}}{\varepsilon_2^2}\right) = O\left(\frac{nB_{\max}\hat{d}_{\max}^{q-1}L^{*2}}{\varepsilon_2^2}\right). \end{aligned}\tag{32}$$

3. Bias of Phase 2 (PD Dropping). Bias arises solely from dropping genuine k^*-stars when $|\mathcal{KS}_u| > L^*$. Let $\rho_u = \min(1, L^*/|\mathcal{KS}_u|)$. A q-clique survives *PD* if its constituent structures are retained, with probability $\rho_u^{\binom{q}{2}}$. By Bernoulli's inequality $(1 - x^c \le c(1-x))$, the squared bias is:

$$\begin{aligned}\mathrm{Bias}_{\mathrm{drop}}^2 &= \left(\sum_{u:|\mathcal{KS}_u|>L^*} f^q(u)(1-\rho_u^{\binom{q}{2}})\right)^2\\ &\le n\sum_{u:|\mathcal{KS}_u|>L^*}(\hat{d}_{\max}^{q-1})^2(1-\frac{L^*}{|\mathcal{KS}_u|})^2. \end{aligned}\tag{33}$$

Applying tail bounds for structural degrees, we obtain the exponential and algebraic forms:

$$\mathrm{Bias}_{\mathrm{drop}}^2 = O\left(\min\left(n\hat{d}_{\max}^{q-1}e^{-c\hat{d}_{\max}^{q-2}}, \frac{n^2\hat{d}_{\max}^{2q-4}}{L^*}\right)\right). \tag{34}$$

By aggregating these terms, we arrive at the total ℓ_2-error bound, as formally stated in Eq. (28). □

6 Experiments

Table 2: Dataset statistics.

Dataset	Nodes	Edges	$\langle d\rangle$	Gini	Sparsity	Type
IMDB [21]	896K	57.1M	127	0.71	1.4e−4	Bipartite
DBLP [36]	317K	1.0M	6.6	0.53	2.1e−5	Collab
MIT [41]	1.9K	3.5K	3.6	0.41	1.8e−3	Temporal
Orkut [36]	3.1M	117M	76	0.55	2.5e−5	Social

6.1 Experimental Setup

Datasets. We used four public graphs spanning diverse sizes, densities, and degree heterogeneity. Table 2 summarizes their statistics: nodes, edges, average degree, Gini coefficient [7] (degree inequality) and sparsity [36].

Experiment Design. We evaluate SPCC on: (1) Overall performance comparison with state-of-the-art solutions across varying privacy budgets; (2) Evaluation on large cliques using dense networks; (3) Analysis of optimal k-star size selection for different clique sizes; (4) *AGP* ablation study and (5) Theoretical error verification.

Competitors. EPCC (Sec. 3), Local2Rounds△ [31], ARRTwoNS△ [32], TDP-VC [26], EdgeOrient△ [42].

Evaluation Metrics. To provide a scale-invariant assessment of estimation accuracy across diverse graph topologies and clique orders, we adopt the Mean Relative Error (MRE) as our primary performance metric. It is defined as:

$$\mathrm{MRE} = \frac{1}{R}\sum_{r=1}^{R}\frac{|f^q(G) - \hat{f}_{(r)}^q(G)|}{f^q(G)}, \tag{35}$$

where $f^q(G)$ denotes the ground-truth q-clique count, $\hat{f}_{(r)}^q(G)$ is the estimate obtained in the r-th independent run, and $R = 10$ is the number of repetitions.

6.2 Overall Results

We benchmark SPCC against EPCC (Section 3)[2] and four state-of-the-art competitors for triangle counting ($q = 3$) across four real-world graphs with $\varepsilon \in [0.5, 4.0]$. SPCC uses optimal k (Eq. 20) and L (Eq. 22) with $\varepsilon_1 = \varepsilon_2 = \varepsilon/2$. Baselines are tuned per their original papers. Fig. 3 shows relative error (log-scale).

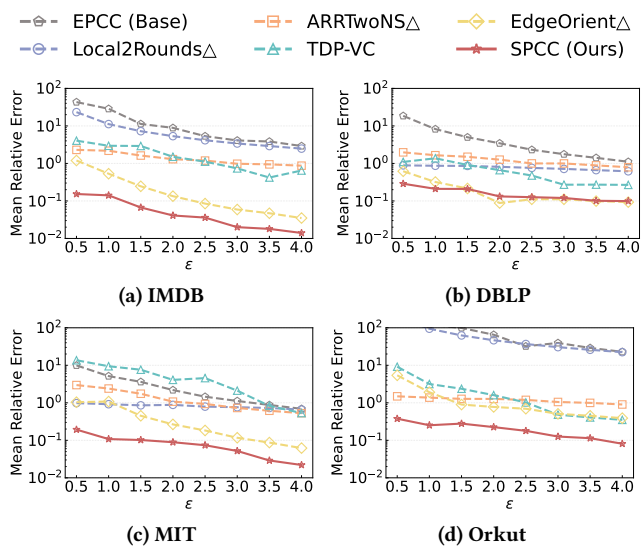


Figure 3: Compare performance with other baselines.

Observations. SPCC achieves the lowest empirical error across most settings. EPCC's variance explosion stems from the multiplicative accumulation of noise across $\binom{n}{3}$ potential triangles. Local2Rounds△ and ARRTwoNS△ perform reasonably on dense graphs (MIT) but degrade on sparse ones (DBLP, Orkut), where false positives overwhelm true signals. Although EdgeOrient△ improves upon these baselines by leveraging graph degeneracy to limit sensitivity, it still lags behind SPCC on massive or dense graphs such as Orkut and IMDB.

SPCC's advantage is most pronounced in heavy-tailed graphs (IMDB): while edge-level perturbation suffers high variance from high-degree nodes, our k-star grouping confines them to smaller domains, effectively allocating more privacy budget.

[2]For EPCC, which theoretically requires $O(n^q)$ enumeration, we use unbiased sampling of 10^5 q-node subsets to obtain feasible runtimes.

ARRTwoNS△ shows diminishing returns with larger ε, while SPCC maintains robust utility gains, widening the gap at higher privacy budgets.

Beyond utility, Table 3 reports the per-user communication overhead and end-to-end runtime across all evaluated mechanisms. The substantial communication in Local2Rounds△ is a direct consequence of the $O(n^2)$ downlink required to broadcast the noisy adjacency matrix. While ARRTwoNS△ and TDP-VC employ sampling to reduce bandwidth, they still suffer from heavy computational overhead due to the simulation of localized subgraph intersections. EdgeOrient△'s efficiency is highly dependent on graph degeneracy; its iterative peeling process for orientation is efficient on power-law graphs like Orkut but becomes a bottleneck on dense topologies like IMDB. EPCC achieves minimal communication by fragmenting data into isolated edges, this extreme sparsity destroys the cohesive structural signals vital for higher-order clique counting, severely degrading utility. By confining the search to a candidate-driven space supported by k-stars, SPCC ensures practical scalability, demonstrating highly competitive runtime and communication overhead on large-scale graphs.

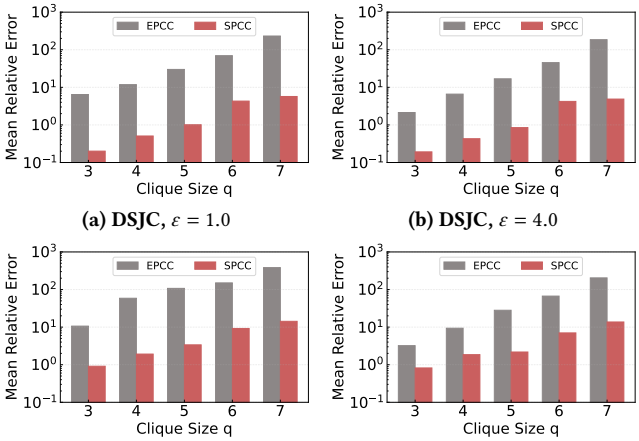
Table 3: Communication Overhead and End-to-End Runtime.

Method	MIT	DBLP	IMDB	Orkut
Communication Overhead (B)				
Local2Rounds△	2.20e5	5.00e9	4.60e10	5.89e11
ARRTwoNS△	3.97e2	1.90e7	1.60e8	2.00e9
TDP-VC	4.60e5	2.10e10	1.82e11	6.00e11
EdgeOrient△	1.40e2	1.90e4	5.40e4	1.89e5
EPCC (Base)	2.37e2	4.00e4	1.12e5	3.87e5
SPCC (Ours)	2.60e4	5.45e5	1.20e7	5.00e9
End-to-End Runtime (s)				
Local2Rounds△	1.00e-2	1.11e2	1.98e3	1.19e4
ARRTwoNS△	1.00e-3	9.50e-1	8.03e0	1.06e2
TDP-VC	1.50e-1	1.25e1	3.24e4	7.60e4
EdgeOrient△	2.00e-1	1.21e3	9.76e3	1.44e4
EPCC (Base)	2.00e0	1.25e3	6.42e4	7.20e4
SPCC (Ours)	5.00e-1	1.00e1	9.00e2	1.80e3

6.3 Evaluation on Large Cliques

We evaluate SPCC on larger cliques ($q \in [3, 7]$). We use two dense benchmarks: (1) DSJC1000.1, a classic random graph ($n = 1,000$, $m = 99,258$, density ≈ 0.2) [33]; and (2) a synthetic Barabási-Albert (BA) graph [3] with $n = 10,000$ and attachment parameter $m = 20$. They can provide sufficient ground-truth signals for evaluating higher-order structural estimation due to their relatively high clique densities. We compare SPCC against EPCC for $q = 3$ to 7 under privacy budgets $\varepsilon = 1.0$ and 4.0. Fig. 4 presents the results.

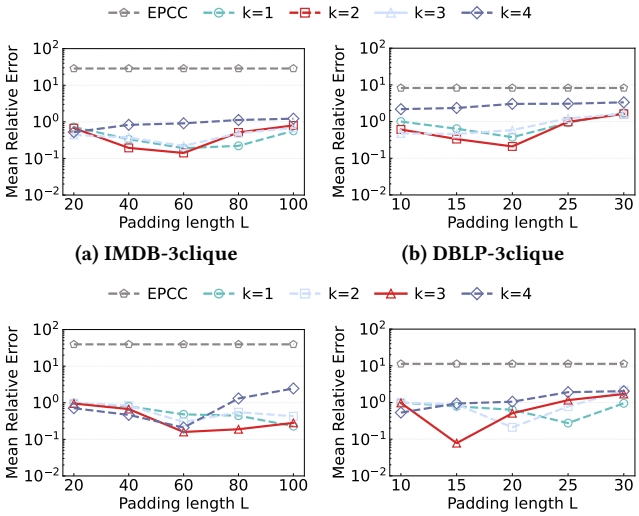
Observations. As expected, EPCC's performance deteriorates significantly as q increases, confirming the severe variance scaling inherent to edge-level LDP models for higher-order cliques. Even in dense benchmarks, multiplicative noise from independent edges generates an excessive number of spurious cliques that mask true signals, rendering EPCC impractical for $q \geq 5$ on the BA graph ($\varepsilon = 1.0$).



(a) DSJC, $\varepsilon = 1.0$ (b) DSJC, $\varepsilon = 4.0$ (c) BA, $\varepsilon = 1.0$ (d) BA, $\varepsilon = 4.0$

Figure 4: Performance of SPCC for different clique sizes.

In contrast, SPCC maintains a stable, low-error profile across all q. This robustness is driven by two factors: (i) k-star perturbation preserves local topological correlations that are typically lost in edge-level methods, and (ii) Targeted Structural Estimation restricts the search space to localized neighborhoods instead of all global combinations, effectively mitigating the variance explosion. And Higher errors on the BA graph compared to DSJC1000.1 stem from its structural heterogeneity; power-law hubs require more aggressive Padding-and-Dropping (*PD*).



(a) IMDB-3clique (b) DBLP-3clique (c) IMDB-4clique (d) DBLP-4clique

Figure 5: Comparison of different k-star perturbation.

6.4 Impact of Varying k-star Representation

As theoretically established in Section 4.6, we show that setting $k = q - 1$ for counting q-cliques optimally preserves clique-relevant

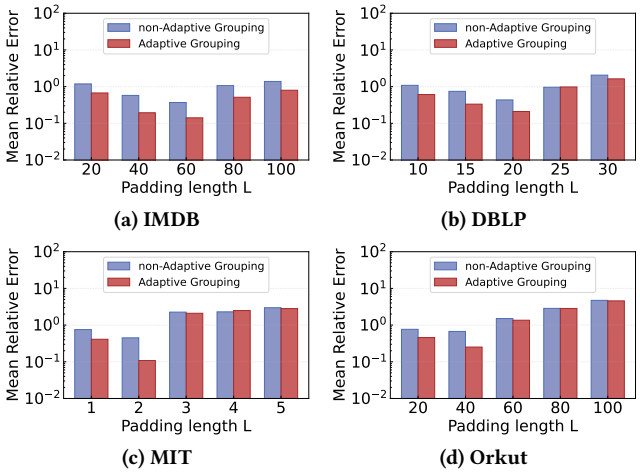
information while controlling sensitivity. We validate this by measuring relative error for 3-clique and 4-clique counting under different k settings on datasets IMDB and DBLP, with EPCC as comparison. Fig. 5 presents the results.

Observations. We observe that $(q-1)$-star configuration achieves optimal performance for counting q-cliques, i.e., $k = 2$ for 3-cliques and $k = 3$ for 4-cliques consistently yield the lowest errors. It validates our theoretical analysis.

We further observe that optimal L depends on k because different k values yield different information density. Large k (e.g., $k = 4$ for 3-clique) introduces extraneous edges, flattening error curves. Conversely, small k suffers more at small L due to information loss.

6.5 Ablation Study on Adaptive Grouping Perturbation (*AGP*)

To validate the effectiveness of the proposed *AGP* mechanism, we conduct an ablation study comparing *Adaptive Grouping* (Group using our *AGP* mechanism) against a naive *non-Adaptive Grouping* (naive group by degree) under a fixed privacy budget $\varepsilon = 1.0$ for triangle counting ($q = 3$). Fig. 6 shows the relative error reduction across all tested graphs and padding lengths.



(a) IMDB (b) DBLP (c) MIT (d) Orkut

Figure 6: Effectiveness of adaptive grouping perturbation.

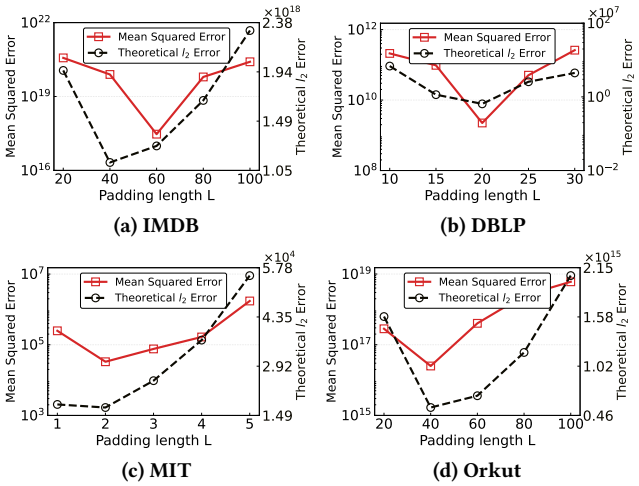
Observations. The *AGP* mechanism reduces relative error by up to 65%, with the most significant gains in graphs having skewed degree distributions (e.g., IMDB). By partitioning high-degree nodes into finer groups, *AGP* allocates them more privacy budget, thereby reducing noise in structurally important regions.

This advantage persists across all padding length settings. For a large L, the smaller privacy budget allocated for each k-star leads to larger noise, so *AGP* will reduce the bucket size to control the overall perturbation error. In contrast, for a small L, which leads to local structural information loss, *AGP* will increase the bucket size to mitigate this influence. Such consistent performance confirms *AGP*'s adaptability to diverse graph structures and operational conditions.

6.6 Theoretical Analysis Validation

In this subsection, we shift our focus from relative utility to the mathematical rigor of our framework. To maintain consistency with the theoretical l_2-error analysis derived in Section 5, we employ Mean Squared Error (MSE) as the empirical metric. This allows for a direct, second-order comparison between our analytical bounds and empirical outcomes, facilitating the validation of our parameter optimization strategies.

6.6.1 Validation of Optimal Padding Length L. We validate the theoretical optimal L (from minimizing Eq. 22) against empirical results. For $\varepsilon = 1.0$ and $q = 3$, Fig. 7 compares the theoretical and empirical error curves across L values.



(a) IMDB (b) DBLP (c) MIT (d) Orkut

Figure 7: Empirical validation of theoretical optimal L.

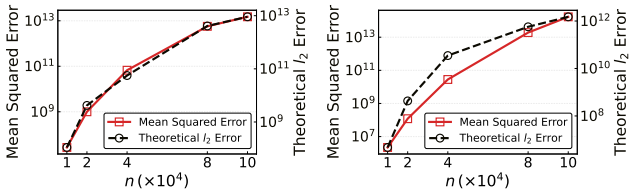
Observations. All datasets exhibit a consistent U-shaped error trend with L, confirming the existence of an optimal length. Small L causes dropping loss; large L introduces padding noise.

Optimal L adapts to graph structure: dense graphs (IMDB) require larger L, while sparse graphs (DBLP, Orkut) degrade sharply with over-padding. The theoretical minima align closely with empirical results, with minor deviations (<5%) due to enumeration complexity.

6.6.2 Validation of the total Theoretical Error. We empirically validate our theoretical error analysis (Eq. 5.2) by comparing its trend against empirical error using sampled subgraphs from the IMDB and Orkut datasets. These large-scale graphs provide sufficient nodes to support stable sampling across a wide range of subgraph sizes.

To mitigate sampling variance, we generate a sequence of induced subgraphs $G_n = (V_n, E_n)$ by uniformly sampling n nodes from the original graph, where n ranges from 10,000 to 100,000. Each sampling process is repeated R times, and we report the average error over these replications, with

$$R = \begin{cases} 100, & 10{,}000 \le n \le 40{,}000, \\ 10, & \text{otherwise.} \end{cases} \tag{36}$$



(a) IMDB (b) Orkut

Figure 8: Verification of the total theoretical error.

Observations. According to the results shown in Fig. 8, we observe that both theoretical error and empirical error increase monotonically with node count n, following a power-law trend spanning four orders of magnitude (10^9 to 10^{13}) as n grows from 2×10^4 to 10^5.

A systematic deviation exists: for small n, empirical variance falls below theory due to undersampled high-degree nodes increasing average group size; for large n, empirical variance exceeds theory as more high-degree nodes — belonging to smaller groups and contributing quadratically to variance — are included.

7 Related Work

The problem of subgraph counting under differential privacy [8, 9, 30, 34, 50] has been extensively explored. Under CDP model, a large body of work has focused on releasing noisy counts of small subgraphs, including triangles and stars [4, 13, 14, 34, 58], cliques [9, 39], 4-cycles and 2-length paths [46]. These methods typically rely on a trusted server and employ techniques such as smooth sensitivity and graph projection to manage the global sensitivity of subgraph queries.

In contrast, under LDP model, the problem becomes significantly more challenging due to higher noise variance and limited structural visibility. Existing LDP research has expanded to count various subgraphs: butterflies in bipartite graphs [25], k-stars [28] and triangles [18, 26, 31, 32, 38, 47, 53], as well as cycles [27, 29]. It is evident that clique counting (for $q > 3$) remains largely unexplored in the LDP setting. Current LDP clique counting focuses mainly on triangles and follows two methodological lines: *Direct Estimation via Structure Collection* and *Indirect Estimation from Local Statistics*.

7.1 Direct Estimation via Structure Collection

The most fundamental approach to triangle counting under LDP involves collecting locally perturbed information from users and reconstructing a noisy graph at the server side, followed by debiasing techniques to estimate global triangle counts. For example, both Eden *et al.* [18] and Saladi *et al.* [47] employ Asymmetric Randomized Response (ARR) to perturb adjacency bits. Users upload their perturbed adjacency lists along with noisy degree information, and the server performs graph down-projection and debiased estimation to infer global counts. Mundra *et al.* [42] utilizes privatized k-core decomposition to derive a low out-degree orientation. By integrating this orientation with randomized response, the algorithm shifts the error bound dependency from total edges to graph degeneracy , achieving tighter theoretical guarantees.

Some work extends users' local views for direct deterministic counting of target structures, followed by noise calibration via sensitivity analysis (e.g., local or tight global sensitivity bounds) and perturbation before aggregation. For example, He et al. [25] propose a multi-round interactive framework where nodes exchange noisy edge information to extend local views; improved by [26] via node sampling and dynamic privacy budgeting in TDP-VC. Liu et al. [38] assume a 2-hop local view for direct triangle counting using controlled sensitivity and sampling. Similarly, Sun et al. [50] extend local knowledge to compute ego-centered subgraph counts including three-hop paths and cliques. However, such strong visibility assumptions are often impractical in real social networks and sometimes incur multi-round interactions, high complexity, and substantial overhead for large graphs.

7.2 Indirect Estimation from Local Statistics

Some methods avoid direct structure counting by collecting local subgraph statistics and using probabilistic models with unbiased estimators. Ye et al. [55] propose LF-GDPR, which collects perturbed adjacency vectors and node degrees, then infers graph metrics via polynomial mapping—general but potentially increasing variance. Similarly, Imola et al. [31] use local statistics in probabilistic models for triangle estimation, while [32] introduce ARR with edge sampling to reduce communication. However, these methods are noise-sensitive, especially in sparse graphs where debiasing amplifies variance and structural discrepancies cause significant errors.

In summary, accurate subgraph counting under LDP faces several limitations: (1) existing methods remain constrained by the edge-level perturbation paradigm, which introduces substantial noise leading to structural distortion and persistently high estimation variance; (2) prohibitive communication and computational overhead for large-scale graphs; (3) a one-size-fits-all privacy model that fails to account for node heterogeneity in real-world networks; and (4) a narrow focus on triangles with limited extensions to larger cliques or complex motifs.

To address these challenges, we propose SPCC, which perturbs k-stars (instead of edges) to preserve richer local structures. It enables accurate and scalable counting of arbitrary q-cliques under rigorous Edge-LDP.

8 Conclusion

We introduce SPCC, a novel LDP framework that enables accurate and scalable clique counting for arbitrary clique sizes by innovating the data representation and perturbation unit from edges to cohesive k-stars. This structural shift fundamentally addresses the limitations of existing edge-based approaches, which suffer from excessive noise and poor preservation of clique connectivity patterns. SPCC integrates two key mechanisms: *Padding-and-Dropping (PD)* for degree concealment and bias control, and *Adaptive Grouping Perturbation (AGP)* for domain reduction and structural signal amplification. Together, they achieve an optimal balance between privacy protection and counting utility, preserving local topological correlations that are essential for accurate clique reconstruction.

SPCC offers significant advantages: it achieves substantially lower estimation error across different graph types and privacy regimes, reduces communication and computation complexity and naturally extends to higher-order cliques without prohibitive overhead. Experimental results confirm its superiority over state-of-the-art LDP methods.

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A Open Science

Our code is available at https://anonymous.4open.science/status/SPCC-637B. The datasets used in our experiments:

IMDB (https://mozart.diei.unipg.it/gdcontest/contest2005/gdevolving2005.html)

DBLP (https://snap.stanford.edu/data/com-DBLP.html)

MIT (https://chrsmrrs.github.io/datasets/docs/datasets/)

Orkut (https://snap.stanford.edu/data/com-Orkut.html)

DSJC (https://mat.tepper.cmu.edu/COLOR/instances.html).